

Given: Due:	Further Maths (Statistics) HW 26	Name:
----------------	---	-------

Chapter 6: Chi Square testing (contingency tables)

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)
----------	---------	---------	---------	---------	---------

Useful resources: Your notes from recent lessons & your text book

Although most Complex arithmetic can be done on your calculator, show all steps to reach the answer & use calculator for checking results only.

Questions - do these on lined paper

Q1. A university foreign language department carried out a survey of prospective students to find out which of three languages they were most interested in studying. A random sample of 150 prospective students gave the following results.

		Language		
		French	Spanish	Mandarin
Gender	Male	23	22	20
	Female	38	32	15

A test is carried out at the 1% level of significance to determine whether or not there is an association between gender and choice of language.

- State the null hypothesis for this test. (1)
- Show that the expected frequency for females choosing Spanish is 30.6 (1)
- Calculate the test statistic for this test, stating the expected frequencies you have used. (3)
- State whether or not the null hypothesis is rejected. Justify your answer. (2)
- Explain whether or not the null hypothesis would be rejected if the test was carried out at the 10% level of significance. (1)

Q2. John thinks that a person's eye colour is related to their hair colour. He takes a random sample of 600 people and records their eye and hair colours. The results are shown in Table 1.

Table 1

		Hair colour				
		Black	Brown	Red	Blonde	Total
Eye colour	Brown	45	125	15	58	243
	Blue	34	90	10	58	192
	Hazel	20	38	16	26	100
	Green	6	29	7	23	65
	Total	105	282	48	165	600

John carries out a χ^2 test in order to test whether eye colour and hair colour are related. He calculates the expected frequencies shown in Table 2.

Table 2

		Hair colour			
		Black	Brown	Red	Blonde
Eye colour	Brown	42.5	114.2	19.4	66.8
	Blue	33.6	90.2	15.4	52.8
	Hazel	17.5	47	8	27.5
	Green	11.4	30.6	5.2	17.9

- (a) Show how the value 47 in Table 2 has been calculated. (1)
- (b) Write down the number of degrees of freedom John should use in this χ^2 test. (1)
- Given that the value of the χ^2 statistic is 20.6, to 3 significant figures,
- (c) find the smallest value of α for which the null hypothesis will be rejected at the $\alpha\%$ level of significance. (1)
- (d) Use the data from Table 1 to test at the 5% level of significance whether or not the proportions of people in the population with black, brown, red and blonde hair are in the ratio 2:6:1:3
State your hypotheses clearly. (9)
- (Total 12 marks)**

Q3. A leisure club offers a choice of one of three activities to its 150 members on a Tuesday evening. The manager believes that there may be an association between the choice of activity and the age of the member and collected the following data.

Activity	Badminton	Bowls	Snooker
Age a years			
$a < 20$	9	3	3
$20 \leq a < 40$	10	10	14
$40 \leq a < 50$	16	15	5
$50 \leq a < 60$	15	13	11
$a \geq 60$	4	19	3

- (a) Write down suitable hypotheses for a test of the manager's belief. (1)
- The manager calculated expected frequencies to use in the test.
- (b) Calculate the expected frequency of members aged 60 or over who choose snooker, used by the manager. (1)
- (c) Explain why there are 6 degrees of freedom used in this test. (2)
- The test statistic used to test the manager's belief is 19.583
- (d) Using a 5% level of significance, complete the test of the manager's belief. (2)
- (Total for question = 6 marks)**

END OF HOMEWORK

Total = 26 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.